

Homework Set 1, CPSC 6430/4430

LastName, FirstName

Due 02/22/2026, 11:59PM EST

Simple Linear Regression

Consider simple linear regression model: $y = ax + b$. Given $\{\{x_1, y_1\}, \{x_2, y_2\}, \dots, \{x_m, y_m\}\}$, please derive the optimal solution is given by:

$$a^* = \frac{\sum_{i=1}^m (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^m (x_i - \bar{x})^2}, \quad (1)$$
$$b^* = \bar{y} - a^* \bar{x},$$

where $\bar{x}$ and $\bar{y}$ represent the mean value of x and y respectively.

Linear Regression in Astronomy

Least Squares was first used by Gauss to predict the orbit of Ceres (the first discovered asteroid/dwarf planet, found in 1801). Kepler's third law states that: The ratio of the square of an object's orbital period with the cube of the semi-major axis of its orbit is the same for all objects orbiting the same primary. By symbol: $T^2 \propto a^3$ where T is the object's orbital period and a is the semi-major axis of its orbit. Please follow these steps to predict the T for Uranus and Neptune.

- Assume you were a researcher before Kepler and you didn't know the exact order between T and a . Let's simply assume $T \propto a^k$, then $T = c * a^k$ where c is a constant and k is to be determined. Thus $\log T = k \log a + b$ where $b = \log c$.
- We can find T and a for Mercury, Venus, Earth, Mars and Jupiter from https://en.wikipedia.org/wiki/Kepler%27s_laws_of_planetary_motion, by making use of linear regression (input $\log a$ and output $\log T$), we can estimate k and b accordingly.
- Extract a for Uranus and Neptune, by using the model we determine above, please predict T and compare with the ground-truth data in the Wikipedia link.
- Please plot in one figure for all planets where x -axis is $\log a$ while y -axis is $\log T$.

Projection Matrix

Consider linear regression problem:

$$\min_{\mathbf{x}} \frac{1}{2} \|\mathbf{Ax} - \mathbf{y}\|_2^2 \quad (2)$$

we say the optimal solution is $\mathbf{x}^* = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{y}$, which states that any $\mathbf{y}$ projected to space $\mathbf{A}$ is given by $\hat{\mathbf{y}} = \mathbf{Ax}^* = \mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{y}$, and we define $\mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$ as the *projection matrix* $\mathcal{P}$. Please prove that:

- $\mathcal{P}^n = \mathcal{P}$ for any n which is positive integer
- Eigenvalue of $\mathcal{P}$ is either 0 or 1
- $\text{trace}(\mathcal{P}) = \text{rank}(\mathcal{P})$

After you have proved all the above, please use Python/Matlab to randomly generate $\mathbf{A}$ (more rows than columns, say $\mathbf{A} \in \mathbb{R}^{10 \times 5}$) and manually verify the correctness of the above conclusions.

Polynomial Regression

Assume you have 5 observation samples $\{\{1, 1\}, \{2, 3\}, \{3, 9\}, \{4, 15\}, \{5, 24\}\}$, now please use a Polynomial regression (order 3) model to approximate the samples, please plot the scatter of samples and then plot the function of the regression model. You are expected to output an image similar to the following:

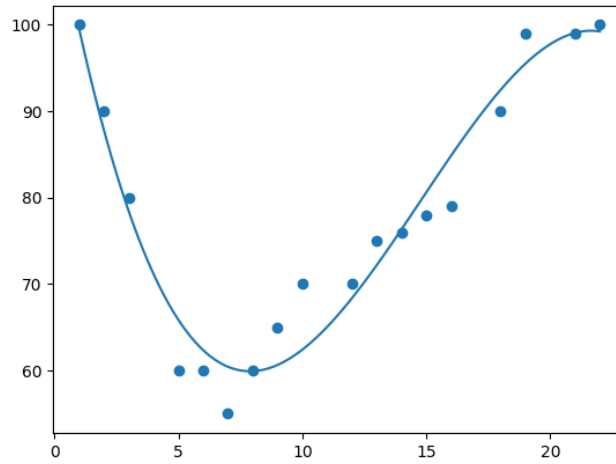


Figure 1: Fitting with Polynomial Regression.

Multiple Linear Regression and Variants for Predicting MPG

Please find the data here, now follow the steps below:

- Extract column ‘cylinders’, ‘displacement’, ‘horsepower’, ‘weight’ and ‘acceleration’ as the input features and column ‘mpg’ as the response ($\mathbf{y}$).
- Randomly split the data into training and testing subsets (at a ratio of 4:1)
- Normalize (via *StandardScaler*) the training samples.
- Use a multiple linear regression model to fit the training data.
- Make the prediction for testing data and calculate the *Mean Squared Error*. (reminder: you should normalize testing data via the *StandardScaler* earlier)
- Fit a **ridge regression** model on the training data.
 - Consider a grid of regularization parameters $\lambda \in \{\lambda_1, \lambda_2, \dots, \lambda_K\}$.
 - For each λ , train the ridge regression model using the normalized training data.
 - Evaluate the model by computing the Mean Squared Error (MSE) on the testing data.
 - Report the value of λ that yields the lowest testing MSE and the corresponding MSE.
- Fit a **lasso regression** model on the training data.
 - Use the same grid of regularization parameters λ as in the ridge regression.
 - Train the lasso regression model for each λ using the normalized training data.
 - Evaluate the testing MSE for each λ .
 - Identify the best λ based on the lowest testing MSE.
 - Report the selected model coefficients and comment on sparsity.
- Compare the performance of **multiple linear regression**, **ridge regression**, and **lasso regression**.
 - Compare their testing MSEs.
 - Discuss the effect of regularization on prediction accuracy.
 - Discuss differences in coefficient magnitudes and feature selection.